What more Samples can do?

Format

Set up independence for the board, then maximum likelihood, Bayes' Rule, and MAP in turn, Markov

Builds on

Lecture 8's chi-square previews — goodness-of-fit closed, independence left as a cliffhanger.

Leads to

Lecture 10 — more concentration inequalities.

RIGHT WHERE WE LEFT OFF

Same table — one side still blank

	Level of significance α							
Two-sided	0.20		0.10		0.05		0.01	
One-sided	0.10		0.05		0.025		0.005	
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10^{-4}	3.84	98.10^{-5}	5.02	16.10^{-5}	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09

$(6-1-(r-1)-((c-1)=2$

Observed

	Python	C++	Java	Total
3rd year	55	30	35	120
4th year	25	35	20	80
Total	80	65	55	200

Expected

	Python	C++	Java	Total
3rd year	48	39	33	120
4th year	32	26	22	80
Total	80	65	55	200

$P(A=3) = \frac{120}{200}$

$P_r(B=\text{python}) = \frac{80}{200}$

$P(r(-(r-1)-((c-1)-1=(r-1)(c-1)$

$$X^2 = \sum_{ij} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$0.6 \quad \frac{65}{200} \times 200$

H_0: Independent.

H_1: student year is not independent of language.

A: Student year.

B: Language used.

$P[A=3 \ \& \ B=\text{python}] = (0.6)(0.4).200$

$= 48$

$P(A=3 \cap B=\text{python}) = P_r(A=3)\ P_r(B=\text{python})$

ONE MORE THING SAMPLES CAN DO

What else can a sample tell you?

So far, a sample has only ever given you a yes-or-no answer — reject H_o, or don't.

Likelihood
L(data | θ) = P(observing this data, if the parameter really were θ)

Maximum likelihood estimate
θ = the value of θ that makes L(data | θ) as large as possible

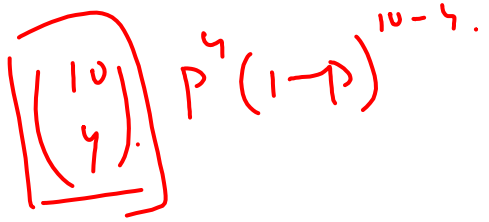
You don't need to know the truth. You pick whichever value makes what you actually saw the least surprising.

WORKED EXAMPLE

Which coin is more believable?

A coin's bias is either 1/3, 1/2, or 2/3. You toss it 10 times and see 4 heads. Which is likeliest?

p (candidate bias)
1/3
1/2
2/3



MLE

$$L(\theta) = \prod_{i=1}^{n} P_r(D_i|\theta)$$

$$\lg L(\theta) = \sum_{i=1}^{n} \lg P_r(D_i|\theta)$$

$$\frac{\partial \lg L(\theta)}{\partial \theta}\Big|_{\theta}$$

Doing the coin example properly

The three-candidate comparison was really a shortcut. Let p range over every value between 0 and 1, and maximize the same likelihood with calculus.

$$L(p) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$\log L(p) = \log\binom{n}{x} + x \log(p) + (n-x)\log(1-p)$$

$$\frac{\partial \log(p)}{\partial p} = \frac{x}{p} - 1 \cdot \frac{n-x}{1-p} = 0 \Rightarrow x(1-p) = (n-x)p$$

$$\Rightarrow p = \frac{x}{n}$$

ANOTHER FAMILIAR ONE

MLE for Geometric RV

Given n independent geometric samples — each X_i the number of trials until the first success. What it its MLE

$$L(p) = \prod_{i=1}^{n} p(1-p)^{x_i - 1}$$

$$= p^n (1-p)^{\sum_{i=1}^{n}(x_i - 1)}$$

$$\log L(p) = n \log(p) + \sum (x_i - 1) \log(1-p)$$

Bayes' Rule, in the vocabulary you'll need

$$P_r(A|B) = \frac{P_r(A \cap B)}{P(B)}$$

$$= \frac{P_r(B|A) \cdot P(A)}{P(B)}$$

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Prior P(H) — what you believed before seeing any data

Likelihood P(E | H) — how probable the data is, if H were true

Marginal P(E) — P(H)P(E|H) + P(¬H)P(E|¬H), just a normalizing constant

Posterior P(H | E) — what you believe after the data

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Same test, two priors, two very different answers

A test is 99% accurate at detecting a condition, and wrongly flags healthy people only 1% of the time. Your friend tests positive. How worried should you be for your friend?

Rare condition — 0.1% prevalence

$P_r(H) = 0.1\%$

$P_r(H|E)$ → tested positive

Having Disease

$$= \frac{(0.99)(0.001)}{(0.99)(0.001) + (0.999)(0.01)}$$

$$= 0.09016$$

Another Independent Tests Positive. Now?

$P_1(H) = 9.1\%$

$$P_r(H|E) = \frac{(.99)(0.09)}{(0.99)(0.09) + (0.91)(0.01)}$$

$$=$$

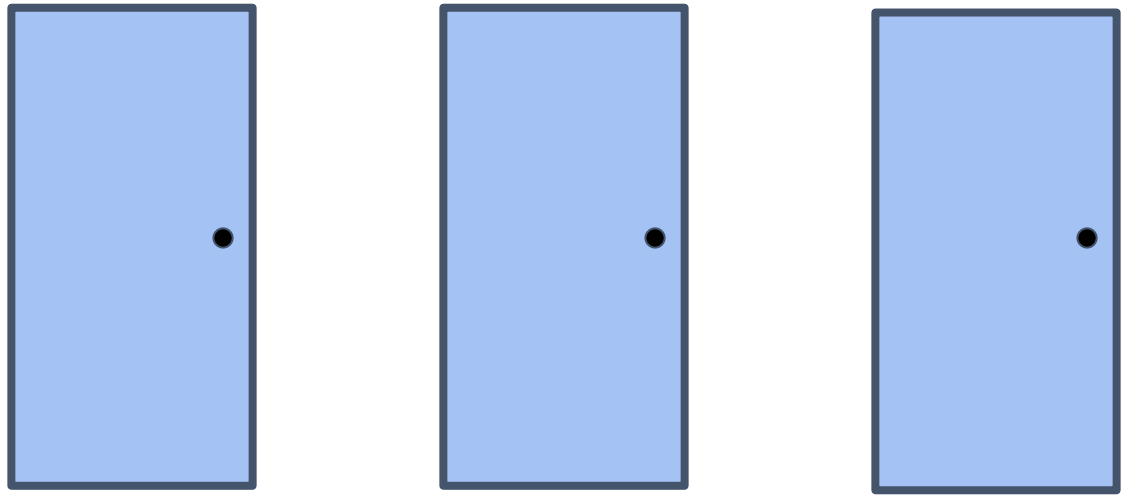
$P_r(H|\sim E)$

$P_r(\sim H|\sim E)$

THE CLASSIC CASE

Monty Hall

There is a car behind one of the three doors

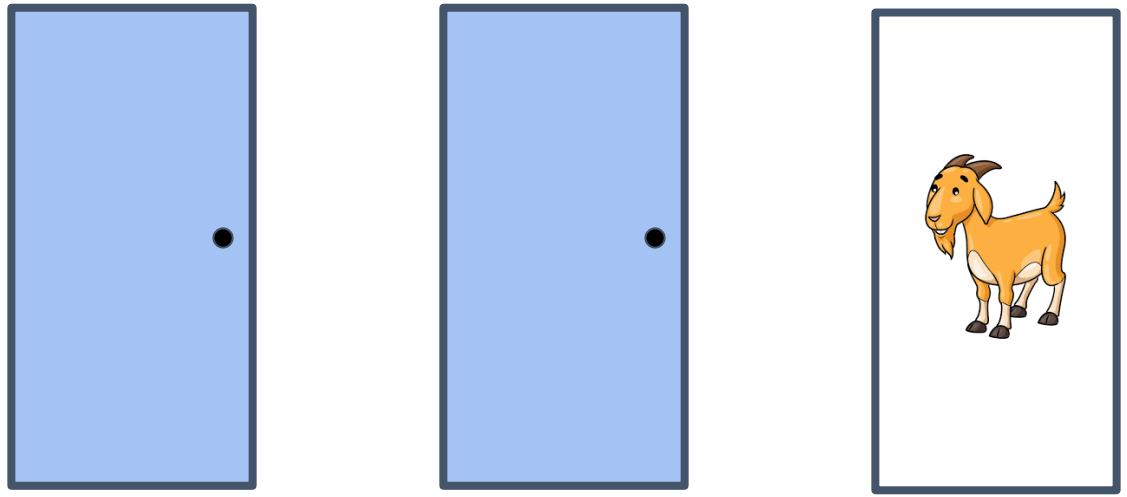


THE CLASSIC CASE

Monty Hall

There is a car behind one of the three doors

You may swap or you may stay with your choice



Door 1 Door 2 Door 3

THE CLASSIC CASE

Monty Hall

There is a car behind one of the three doors



Door 1



Door 2



Door 3

H: Door that has the car (H = 1, 2, 3).
E: Door that Monty reveals (E = 1, 2, 3)
Let Door1 is selected and Door3 revealed.
Compute, Prior, Likelihood, Marginal to get Posterior.

$P_r(H=1|E=3)$

$P_r(E=3|H=1) = 1/2$

$P_r(E=3|H=2) = 1$

$P_r(E=3|H=3) = 0$

$P_r(H=1,2,3) = \frac{1}{3}$

$= \frac{1/2 \cdot 1/3}{(\quad)}$

$P_r(H=2|E=3)$

$\frac{1 \cdot 1/3}{(\quad)}$